

Fibonacci number or sequence

$$\{1, 1, 2, 3, 5, 8, 13, 21, 34, \dots\}$$

f_n

$$f_n = f_{n-1} + f_{n-2}$$

$$f_0 = 1, \quad f_1 = 1$$

$$\begin{aligned} f_2 &= f_{2-1} + f_{2-2} \\ &= f_1 + f_0 \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

$$\begin{aligned} f_3 &= f_{3-1} + f_{3-2} \\ &= f_2 + f_1 \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

$$\begin{aligned} f_4 &= f_{4-1} + f_{4-2} \\ &= f_3 + f_2 \\ &= 3 + 2 \\ &= 5 \end{aligned}$$

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Fibonacci Method :-

1) Determine the number of iterations

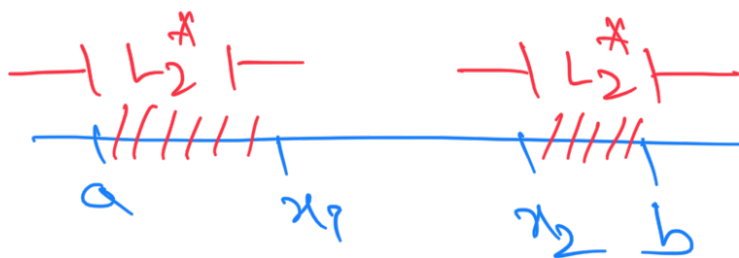
2) Define $L_2^* = \frac{F_{n-2}}{F_n} L_0$

where $L_0 = b - a$ is the length of initial interval of uncertainty

Choose the two initial points x_1 and x_2 be such that

$$x_1 = a + L_2^*$$

$$x_2 = b - L_2^*$$



3) Compute $f(x_1)$, $f(x_2)$ and apply unimodal

Principal

4) The process is repeated again and again

Fibonacci number

f_0 f_1 f_2 f_3 f_4 f_5 f_6 f_7
1 1 2 3 5 8 13 21

Question:- Minimize $f(x) = x^2$ over

$[-5, 15]$ Take $n = 7$

Preserve L and delete R

		$-5 \quad L$		x_1		x_2		$R \quad 15$	
K	$\frac{f_{n-k}}{f_{n-k+1}}$	L	R	x_1	x_2	$f(x_1)$	$f(x_2)$	L/R	
1	$\frac{f_6}{f_7} = \frac{13}{21}$	-5	15	2.6191	7.3809	6.8596	54.4276	L	Preserve L and delete R
2	$\frac{f_5}{f_6} = \frac{8}{13}$	-5	7.3809	-0.2352	2.6191	0.0557	6.8596	L	
3	$\frac{f_4}{f_5} = \frac{5}{8}$	-5	2.6191	-2.1427	-0.2352	4.5911	0.0557	R	
4	$\frac{f_3}{f_4} = \frac{3}{5}$	-2.1427	2.6191	-0.2382	0.7146	0.0557	0.5106	L	
5	$\frac{f_2}{f_3} = \frac{2}{3}$	-2.1427	0.7146	-1.1899	-0.2382	1.4158	0.0557	R	
6	$\frac{f_1}{f_2} = \frac{1}{2}$	-1.1899	0.7146	-0.2382	0.0567	0.0567	0.0557	R	

7	$\frac{1}{1}$	0.7146	-0.2382	X	X	X	X	X
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$$x_{\min} \in [-0.2382, 0.7146]$$

$$\text{Hence } x^* = \frac{-0.2382 + 0.7146}{2}$$

$$= 0.2382$$

$$f(x^*) = 0.05674 \quad \left(\begin{array}{l} \text{min of the} \\ \text{func corresponding} \\ \text{to } [-5, 5] \\ \text{for } n=7 \end{array} \right)$$

H/W

Question 2: Minimize $f(x) = x(x-1.5)$
in $[0, 1]$ within the interval
& uncertainty 0.25 & the
initial interval & uncertainty

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Hint:

Here n is not given

$$L_n \leq 0.25 L_0$$

$$\Rightarrow \frac{L_n}{L_0} \leq 0.25$$

we know $\frac{L_n}{L_0} = \frac{1}{F_n}$

$$\Rightarrow \frac{1}{F_n} \leq 0.25$$

$$\Rightarrow F_n \geq 4$$

So Fibonacci series

F_0	F_1	F_2	F_3	F_4	F_5
1	1	2	3	5	8

Proof Same as above
manner can do

Golden Section Method

The Golden Section is a technique to find out the extremum (max or min)

of a strictly unimodal func.

Method:-

Let initial interval $[a, b]$ be given and the number of iterations be n

1) Set $L = a, R = b, K = 1$ (iteration number)

2) Compute x_1 and x_2 as

$$x_2 = L + 0.618(R - L) \text{ and}$$

$$x_1 = L + R - x_2$$

If $f(x_1) < f(x_2)$ then

$$L_{K+1} = L_K$$

$$R_K \leftarrow x_2; x_2 \leftarrow x_1$$

i.e. Preserve L and
update R with x_2

If $f(x_2) \leq f(x_1)$
then

$$R_{K+1} = R_K$$

$$L_{K+1} \leftarrow x_1; x_1 \leftarrow x_2$$

i.e. Preserve R
and update
 L with x_1

3) Repeat step 2 Until
requirement met

Question: - Minimize $f(x) = x^2$ over $[5, 15]$

Using Golden Section Take $n=7$

K	L	R	x_1	x_2	$f(x_1)$ L	$f(x_2)$ R	L/R
1	-5	15	$x_1 = L + R - x_2$ 2.64	$x_2 = L + 0.618(R-L)$ 7.36	↓ min 6.9696	54.17	L
2	-5	7.36	-0.28	2.64	↓ min 0.0784	6.9696	L
3	-5	2.64	-2.08	-0.28	4.3264	↓ min 0.0784	R
4	-2.08	2.64	-0.28	0.84	↓ min 0.0784	0.7056	L
5	-2.08	0.84	-0.96	-0.28	0.9216	↓ min 0.0784	R
6	-0.96	0.84	-0.28	0.16	0.0784	↓ min 0.0256	R
7	-0.28	0.84	0.16	0.4	0.0256	0.16	

$$x_{\min} \in [-0.28, 0.84]$$

$$\text{Hence } x^* = \frac{-0.28 + 0.84}{2} \\ \approx 0.06$$

$$f(x^*) = (0.06)^2 \\ = 0.0036$$

NOTE:-

Multivariate Function

Minimization by the

method of Steepest descent,

Nelder-Mead Algorithm

Can be taken from
your assigned
ASSIGNMENT

Refer to the books
in Library for ✓
Practice Questions. ✓

Do Prepare Well

ALL THE BEST

